



Report on SOC Alert Management and Incident Response

- **Alert Priority Levels**

- In a SOC environment, thousands of alerts may be generated daily from SIEMs, EDR solutions, IDS/IPS systems, and vulnerability scanners. Not every alert represents the same level of risk. Alert prioritization is therefore essential to ensure that the most dangerous and urgent threats are handled first.

1. Priority Definition

- **Severity Levels**

- SOC environments commonly classify alerts into four categories:

Priority Level	Description	Example
Critical	Immediate threat causing severe impact	Ransomware encryption
High	Serious threat requiring urgent response	Unauthorized admin access
Medium	Suspicious activity with moderate impact	Repeated failed logins
Low	Minor or informational events	User policy violation

1. Critical Alerts

- Critical alerts represent the highest severity level and require immediate incident response actions
- If ransomware starts encrypting files on a company's production server, the SOC immediately classifies the incident as Critical severity because business operations may stop completely and important data may become inaccessible.

2. High Severity Alerts

- High-priority alerts indicate serious threats that may lead to compromise if not handled quickly.



- It includes :
 - unauthorized administrator access
 - malware on critical endpoints
 - privilege escalation attempts
 - exploitation of high-risk vulnerabilities

3. Medium Severity Alert

- Medium-priority alerts represent suspicious or potentially malicious behavior but may not immediately threaten business operations.
- It include :
 - repeated failed login attempts
 - suspicious outbound network connections
 - internal port scanning
 - unusual login times

4. Low Severity Alerts

- Low-priority alerts are informational or low-risk events that generally do not require immediate response.
- It include :
 - blocked phishing emails
 - isolated vulnerability scans
 - policy violations
 - unsuccessful malware execution attempts

2. Assignment Criteria

1. Asset Criticality

- The importance of the affected system significantly influences priority levels.

Asset Type	Priority Impact
Production Server	High/Critical
Database Server	High/Critical
Domain Controller	Critical
Test VM	Medium/Low

2. Exploit Likelihood

- Analysts evaluate how likely a vulnerability or attack can be exploited.
- Exploit likelihood refers to the probability that attackers can successfully exploit a vulnerability or security weakness.



- A vulnerability becomes more dangerous when exploit code is publicly available because attackers can easily use ready-made tools without developing their own exploit.

3. Business Impact

- SOC teams evaluate how the incident affects on business operations, financial systems, customer trust, regulatory compliance.
- Business impact evaluates how much damage the incident could cause to organizational operations, finances, reputation, or compliance requirements.

3. CVSS Scoring System

- The Common Vulnerability Scoring System (CVSS) is an industry-standard framework used to measure vulnerability severity and support risk prioritization.
- CVSS scores range from : 0.0 – 10.0\
- Higher scores represent more severe vulnerabilities.
- One of the most dangerous vulnerabilities in recent years was: CVE-2021-44228 (Log4Shell). This vulnerability received a CVSS score of: 9.8/10

4. NIST SP 800-61 Framework

- National Institute of Standards and Technology provides incident response guidance through: NIST SP 800-61
- The framework recommends classifying incidents based on functional impact, information impact, recoverability, operational disruption.
- **summary**
 - Alert priority levels help SOC analysts classify and respond to security incidents efficiently based on severity, exploit likelihood, business impact, and asset importance. By using frameworks such as CVSS and NIST SP 800-61, organizations can standardize risk assessment and improve incident response processes. Real-world examples such as ransomware attacks and the Log4Shell vulnerability demonstrate the importance of proper alert prioritization in modern cybersecurity operations.

● **Incident Classification**

- Incident classification is the process of identifying, categorizing, and labeling cybersecurity incidents based on their characteristics, attack methods, impact, and behavior. In a Security Operations Center (SOC), proper classification helps analysts understand threats quickly, prioritize response actions, improve communication, and standardize investigations.
- Without proper classification, SOC teams may struggle to identify attack patterns, correlate related incidents, or escalate threats correctly. Standardized classification



frameworks such as MITRE ATT&CK, ENISA Taxonomy, and VERIS help organizations maintain consistency across security operations.

1. Incident Categories

- Security incidents are classified into different categories depending on the type of attack or malicious activity detected. Each category helps analysts understand attacker behavior and choose appropriate response actions.

1. Malware Incidents

- Malware incidents involve malicious software designed to damage systems, steal data, or provide unauthorized access.
- Common malware types include:
 1. ransomware
 2. trojans
 3. spyware
 4. worms
 5. keyloggers
- eg. A workstation infected with ransomware encrypting company files is classified as: Malware Incident – Ransomware

2. Phishing Incidents

- Phishing attacks attempt to trick users into revealing credentials, downloading malware, or clicking malicious links.
- Phishing attacks may occur through emails, fake login pages, SMS messages, social engineering
- Eg. An employee receives a fake Microsoft 365 login email requesting password verification. This classifies into Phishing Incident.
- Phishing emails are commonly mapped to: T1566 – Phishing.

3. DDoS Incidents

- Distributed Denial-of-Service (DDoS) attacks attempt to overwhelm systems, applications, or networks with massive traffic, making services unavailable.
- A company website receives millions of requests from multiple IP addresses causing server downtime it classifies into : DDoS Attack.

4. Insider Threat Incidents

- Insider threats involve malicious or negligent actions performed by trusted employees, contractors, or internal users.
- An employee copies confidential customer data to a personal USB device without authorization. It classifies into Insider Threat.

5. Data Exfiltration Incidents

- Data exfiltration occurs when attackers steal sensitive information from organizational systems.



- Eg. Large volumes of confidential files are uploaded to an unknown external cloud storage service it classifies into : Data Exfiltration.

2. Incident Taxonomy

• MITRE ATT&CK Framework

- The MITRE ATT&CK framework is widely used in SOC environments to classify attacker tactics and techniques.
- MITRE ATT&CK helps to analyse identify attacker behavior,, map incidents to attack stages, improve threat detection, standardize investigations

• Attack techniques :

Credential Dumping : T1003 – Credential Dumping

- This technique involves attackers extracting passwords or hashes from operating systems.

• ENISA Incident Taxonomy

- European Union Agency for Cybersecurity (ENISA) provides a structured taxonomy for incident classification.
- ENISA categories include:malicious code, denial of service, unauthorized access, information gathering, availability issues

• VERIS Framework

- VERIS stands for Vocabulary for Event Recording and Incident Sharing
- VERIS helps organizations record incidents using a standardized structure.
- It focuses on threat actors, attack actions, affected assets, impact analysis.
- VERIS Classification :
 1. threat actor: external attacker
 2. action: phishing
 3. asset: employee email account
 4. impact: credential compromise

3. Contextual Metadata

- Incident classification becomes more effective when analysts enrich incidents with contextual metadata. Metadata provides additional information that supports investigation, threat hunting, and response activities.
- metadata includes timestamps, source IP addresses, destination IP addresses, usernames ,affected systems, malicious file hashes, URLs, process names and IOCs.

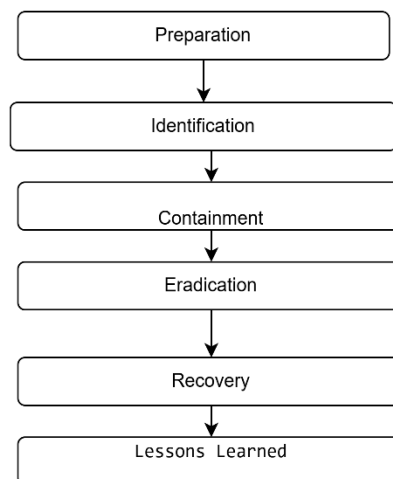


- **Basic Incident Response**

- Incident Response (IR) is a structured process used by organizations to detect, analyze, contain, and recover from cybersecurity incidents. The primary goal of incident response is to minimize damage, reduce recovery time, protect sensitive data, and restore normal business operations as quickly as possible.
- Industry-standard frameworks such as National Institute of Standards and Technology NIST SP 800-61 and the SANS Institute Incident Handler's Handbook provide best practices for handling security incidents effectively.

1. Incident Response Lifecycle

- Each phase plays an important role in managing cybersecurity incidents efficiently.



1. Preparation Phase

- Preparation is the foundation of effective incident response. In this phase, organizations establish the tools, procedures, communication channels, and security controls needed to respond to incidents quickly and effectively.
- Without preparation, SOC teams may struggle during real attacks due to confusion, missing procedures, or lack of coordination.
- A SOC team creates a phishing response playbook containing:
 1. investigation steps
 2. IOC analysis procedures
 3. containment actions
 4. communication templates



2. Identification Phase

- The identification phase begins when suspicious activity or alerts indicate a potential security incident. Analysts investigate alerts to determine whether malicious activity is actually occurring.
- This phase focuses on:
 1. alert validation
 2. event analysis
 3. incident classification
 4. IOC identification
- Alert Triage :Analysts prioritize incidents based on:
 1. severity
 2. business impact
 3. affected systems
 4. exploit likelihood

6. Containment Phase

- Containment focuses on stopping the attack from spreading further while preserving evidence for investigation.
- The goal is to minimize damage without disrupting important business operations unnecessarily.
- System Isolation : Compromised systems may be:
 1. disconnected from networks
 2. isolated using EDR tools
 3. blocked by firewalls

3. Eradication Phase

- Eradication removes the root cause of the incident completely from the environment.
- SOC teams eliminate malware, attacker persistence mechanisms, vulnerabilities, and unauthorized access.
- Malware Removal`:
 1. remove malicious files
 2. delete persistence mechanisms
 3. clean infected systems

4. Recovery Phase

- Recovery focuses on restoring affected systems and services back to normal operations safely.
- SOC teams monitor systems carefully to ensure attackers do not regain access.
- System Restoration :
 1. restore systems from backups
 2. rebuild compromised machines
 3. restart services



5. Lessons Learned

- The lessons learned phase occurs after incident resolution. SOC teams review the incident to identify weaknesses, improve procedures, and prevent future attacks.
- This phase is critical for continuous improvement.
- Post mortems : Teams analyze:
 1. attack timeline
 2. detection effectiveness
 3. response delays
 4. communication gaps

2. Procedures:

• system isolation

- System isolation is a containment procedure used during incident response to prevent a compromised device or system from communicating with other systems on the network. The main goal of isolation is to stop malware spread, attacker movement, data exfiltration, or additional damage while preserving evidence for investigation.
- Without isolation, attackers may continue spreading across the network or steal additional data.
- SOC analysts isolate systems to:
 1. stop malware propagation
 2. prevent lateral movement
 3. block attacker communication
 4. protect sensitive systems
 5. reduce business impact
 6. preserve forensic evidence

• Evidence Preservation

- Evidence preservation ensures that digital evidence remains intact for forensic analysis, legal investigations, and compliance requirements.
- Improper handling may destroy critical evidence.
- Eg. Memory Dumps, File Hashing.

• communication protocols

- Clear communication is essential during incident response.
- SOC teams communicate with:
 1. management
 2. IT teams
 3. legal departments
 4. external vendors
 5. affected users

• SOAR tools

- SOAR stands for: Security Orchestration, Automation, and Response



- SOAR platforms automate repetitive incident response tasks.
- SOAR tools can:
 1. enrich alerts automatically
 2. block malicious IPs
 3. isolate endpoints
 4. create incident tickets
 5. trigger playbooks

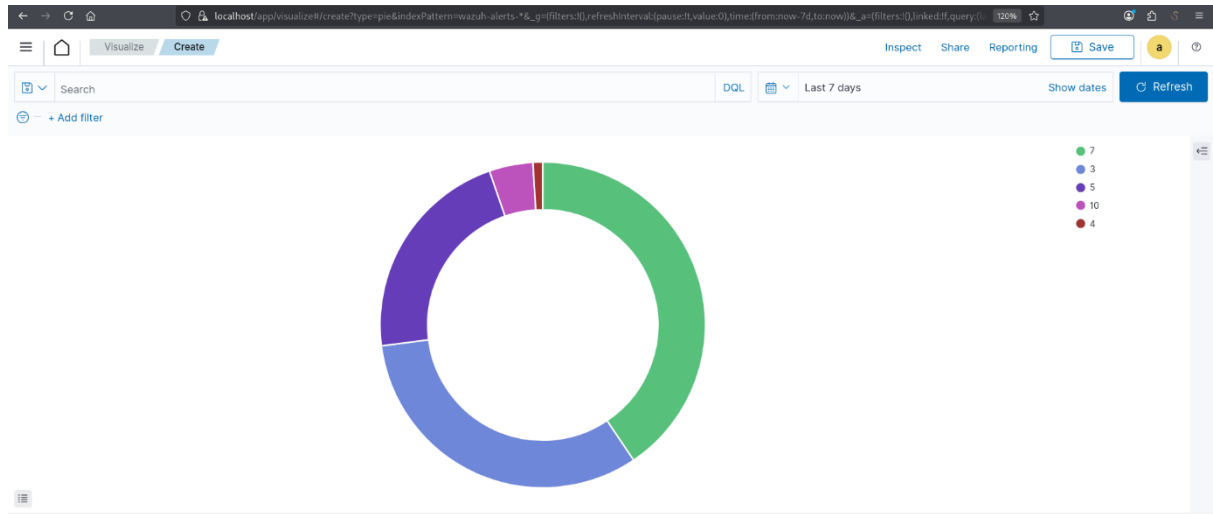
Alert Management Practice

- **Objective**
 - The objective of this practical activity was to understand the process of alert management in a Security Operations Center (SOC) using Wazuh SIEM. The activity focused on generating security alerts, classifying alerts, prioritizing incidents, analyzing events, and documenting response procedures.
- **Alert classification system**

	A	B	C	D	E	F	G	H	I
1	Alert ID	Alert Type	CVSS Score	Asset Criticality	Assigned Priority	MITRE Technique	Description	Status	
2	1	Phishing Email	8.1	Employee Mailbox	High	T1566	Suspicious email link detected	Open	
3	2	SSH Brute Force	6.5	Linux Server	Medium	T1110	Multiple failed SSH login attempts detected	Investigating	
4	3	Log4Shell Exploit	9.8	Production Server	Critical	T1190	Remote code execution attempt identified	Open	
5	4	Port Scan	3.2	Test VM	Low	T1046	Multiple ports scanned from external IP	Closed	
6	5	Malware Execution	8.9	Finance Workstation	High	T1204	Suspicious executable launched by user	Investigating	
7	6	PowerShell Attack	9.1	Domain Controller	Critical	T1059	Encoded PowerShell command executed	Open	
8	7	Unauthorized Admin Login	8.4	Database Server	High	T1078	Admin account logged in from unknown IP	Open	
9	8	DNS Tunneling	7.8	Internal Network	High	T1071	Suspicious DNS traffic pattern detected	Investigating	
10	9	USB Device Detection	4.5	HR Workstation	Medium	T1091	Unauthorized USB storage device connected	Closed	
11	11	Failed MFA Attempts	5.9	VPN Gateway	Medium	T1110	Multiple MFA verification failures	Investigating	
12	12	Suspicious File Download	6.8	Employee Laptop	Medium	T1105	Download from known malicious domain	Open	
13	13	Credential Dumping	9.5	Active Directory Server	Critical	T1003	LSASS memory access detected	Open	
14	14	Web Shell Upload	8.7	Web Server	High	T1505	Unauthorized PHP shell uploaded	Investigating	
15	15	ARP Spoofing Attempt	5.2	Internal LAN	Medium	T1557	Possible man-in-the-middle activity	Open	
16	16	Disabled Antivirus	7.3	Windows Endpoint	High	T1562	Endpoint protection service disabled	Open	
17	17	Excessive File Access	6.1	Shared Storage	Medium	T1530	User accessed large number of files	Investigating	
18	18	SQL Injection Attempt	8.6	Web Application	High	T1190	Malicious SQL query patterns detected	Open	
19	19	Reverse Shell Connection	9.4	Linux Server	Critical	T1059	Outbound reverse shell connection observed	Open	
20	20	Suspicious Remote Desktop Login	7.1	Windows Server	High	T1021	RDP login from unusual geographic location	Investigating	
21									
22									



- **Visualization of a pie chart in wazuh**



- **Drafting a ticket in TheHive**

Title: [Critical] Ransomware Detected on Server-X

Description:

Indicators detected:

- File: crypto_locker.exe
- Source IP: 192.168.1.50
- Multiple suspicious file encryption activities identified.
- Unauthorized process execution observed.

Priority: Critical

Assignee: SOC Analyst

Status: Open

Action Taken:



- Endpoint isolated from network
- Initial IOC investigation started
- Logs preserved for forensic analysis
- Escalated to Tier 2 SOC team

- **Escalation Email**

Dear Tier 2 SOC Team,

A critical ransomware-related alert has been identified on Server-X during SOC monitoring activities. Suspicious encryption behavior and unauthorized process execution were detected.

Indicators of Compromise (IOCs):

- File: crypto_locker.exe
- Source IP: 192.168.1.50

The affected endpoint has been isolated from the network to prevent further spread. Relevant logs and evidence have been preserved for investigation and forensic review.

Please proceed with advanced analysis and containment validation.

Regards,
SOC Analyst

Response Documentation

Creating incident response templates

Phishing Incident Report

- **Executive Summary**

A phishing email containing a suspicious link was detected during SOC monitoring activities. The email attempted to trick users into entering login credentials on a fake login page. Immediate investigation and containment actions were performed to prevent unauthorized access and credential compromise



- **TimeLine**

Time stamp	Activity
2026-07-15 14:00:00	Phishing email detected
2026-07-15 14:20:00	Affected endpoint isolated
2026-07-15 14:30:00	Suspicious URL analyzed
2026-07-15 14:40:00	Memmory dump collected
2026-07-15 14:45:00	IOC inestigation completed
2026-07-15 14:50:00	User password reset intiateted

- **Impact Analysis**

The phishing email targeted internal users and attempted credential theft through a malicious login page. One endpoint interacted with the suspicious URL before detection. No confirmed data exfiltration was identified during the investigation.

- **Remediation Steps**

Isolated affected endpoint from network Blocked malicious URL Reset affected user credentials Performed antivirus and malware scans Preserved logs for forensic analysisConducted IOC review across systems

- **Lessons Learned**

The incident highlighted the importance of user awareness training and rapid phishing detection mechanisms. Improved email filtering policies and enhanced SOC monitoring rules can reduce response time and prevent similar phishing attempts in future operations.



Investigation steps table

Timestamp	Action
2026-03-11 12:00:00	Isolated endpoint
2026-03-11 12:10:00	Verified phishing email headers
2026-03-11 12:20:00	Checked malicious URL reputation
2026-03-11 12:30:00	Collected memory dump
2026-03-11 12:45:00	Preserved system logs
2026-03-11 12:00:00	Reset compromised credentials

Phishing checklist

- ☐ Confirm email headers
- ☐ Verify sender domain
- ☐ Check link reputation using VirusTotal
- ☐ Identify affected users
- ☐ Isolate affected endpoint
- ☐ Preserve logs and evidence
- ☐ Reset compromised passwords
- ☐ Block malicious domains/IPs
- ☐ Update SIEM detection rules
- ☐ Document incident findings

- **Post-mortem summary**

The simulated phishing incident demonstrated the importance of rapid detection, endpoint isolation, and IOC validation during security investigations. Process improvements include stronger email filtering, faster alert escalation, regular phishing awareness training, and improved documentation procedures to enhance future incident response efficiency and reduce potential business impact.



Alert Triage Practice

- Failed SSH login attempts were generated from a separate Windows system targeting the Kali Linux SSH service. The activity produced realistic authentication logs containing the attacker source IP address, enabling practical SIEM monitoring, alert triage, and IOC investigation within the Wazuh environment.
- **Attempting a fake login attempts**

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\akshay>ping 10.186.27.171

Pinging 10.186.27.171 with 32 bytes of data:
Reply from 10.186.27.171: bytes=32 time=33ms TTL=64
Reply from 10.186.27.171: bytes=32 time=1ms TTL=64
Reply from 10.186.27.171: bytes=32 time=2ms TTL=64

Ping statistics for 10.186.27.171:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 33ms, Average = 12ms
Control-C
^C
C:\Users\akshay>ssh fakeuser@10.186.27.171
fakeuser@10.186.27.171's password:
Permission denied, please try again.
fakeuser@10.186.27.171's password:
Permission denied, please try again.
fakeuser@10.186.27.171's password:
fakeuser@10.186.27.171: Permission denied (publickey,password).

C:\Users\akshay>
```

- **Triaging the alert in wazuh**

wazuh. Modules Security events							
Security Alerts							
Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
> May 14, 2026 @ 18:54:24.181	000	kali	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
> May 14, 2026 @ 18:54:24.181	000	kali	T1110	Credential Access	Multiple failed SSH login attempts detected	10	100100
> May 14, 2026 @ 18:54:24.178	000	kali	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
> May 14, 2026 @ 18:54:24.178	000	kali	T1110	Credential Access	Multiple failed SSH login attempts detected	10	100100
> May 14, 2026 @ 18:54:24.178	000	kali	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
> May 14, 2026 @ 18:54:22.178	000	kali	T1110	Credential Access	Multiple failed SSH login attempts detected	10	100100



- Event info

Security events

>	May 14, 2026 @ 18:54:24.181	000	kali	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
<	May 14, 2026 @ 18:54:24.181	000	kali	T1110	Credential Access	Multiple failed SSH login attempts detected	10	100100

Table JSON Rule

@timestamp	2026-05-14T13:24:24.181Z
_id	FJ-oJp4807yey471aX
agent.id	000
agent.name	kali
data.scrip	10.186.27.161
data.srcuser	fakeuser
decoder.name	sshd
decoder.parent	sshd
full_log	2026-05-14T18:54:23.830100+05:30 kali sshd-session(6697): Failed password for invalid user fakeuser from 10.186.27.161 port 60883 ssh2
id	1778765064.75924
input.type	log
location	/var/log/auth.log
manager.name	kali
predecoder.program_name	sshd-session
predecoder.timestamp	2026-05-14T18:54:23.830100+05:30

wazuh.

Modules

Security events

Security Alerts

Time	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
> May 14, 2026 @ 18:54:24.181	000	kali	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
< May 14, 2026 @ 18:54:24.181	000	kali	T1110	Credential Access	Multiple failed SSH login attempts detected	10	100100

Table

JSON

Rule

<

Information

>

View in Rules

ID

Level

File

Path

100100

10

Groups

custom_ssh_bruteforce

<

Details

>

Frequency

Timeframe

if_matched_sid

Same_source_ip

3

120

5710

true

<

Compliance

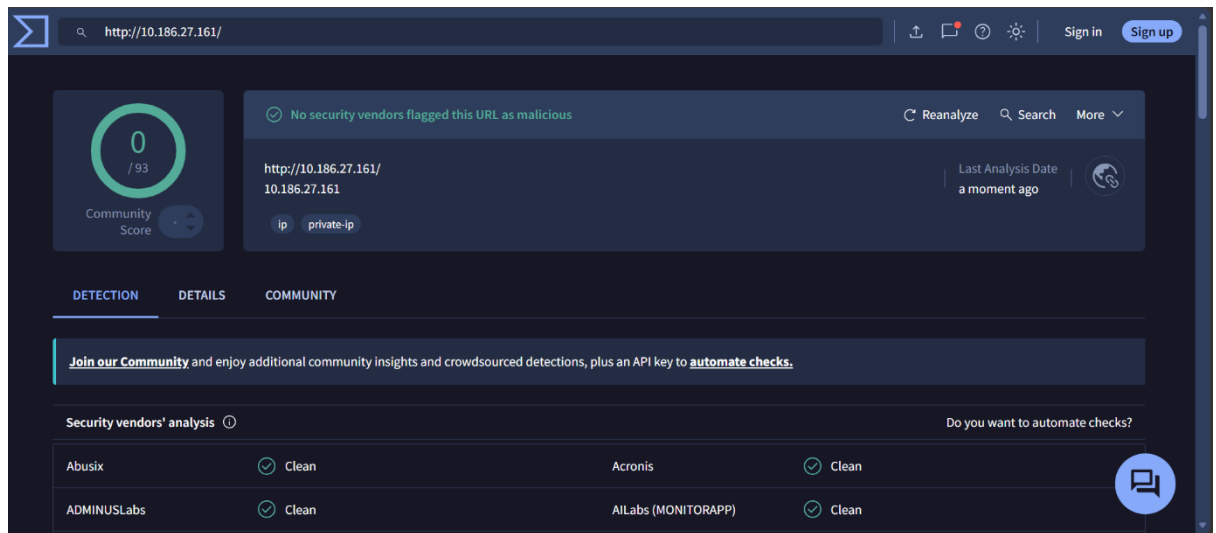
>

MITRE

T1110



- Findings on virustotal(Threat intel validation)



- Triage table

Alert ID	Description	Source IP	Priority	Status
002	Brute-force SSH Attempts	10.187.27.161	Medium	Investigating

- Triage Summary

A simulated brute-force SSH attack was generated using repeated failed login attempts against the localhost SSH service. Wazuh successfully detected and logged the authentication failures. The alert was analyzed and categorized as Medium priority due to repeated unauthorized access attempts. Threat intelligence validation procedures were also reviewed using VirusTotal .



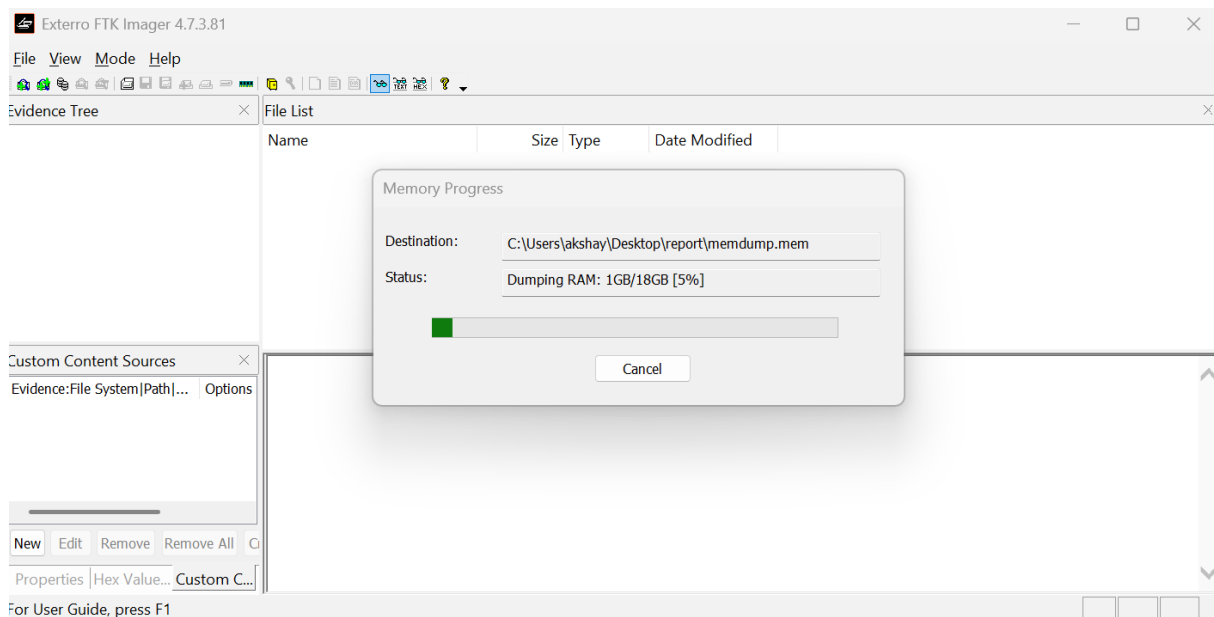
- **Evidence Preservation**

- Evidence preservation is a critical phase in digital forensics and incident response. During a cyberattack or suspicious activity, investigators must preserve logs, system data, network activity, and artifacts without altering their integrity.

- **Collecting network connections (csv format)**

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
	State	AppliedSet	OffloadSta	Caption	Descriptio	ElementN	InstanceC	Communic	DetailedSt	HealthSta	InstallDa	Name	Operating	Operation	PrimarySt	Status	StatusDes	AvailableR	EnabledDe	En
1	Bound		InHost				+++38604+++	++0												2
2	Bound		InHost				+++38582+++	++0												2
3	Bound		InHost				+++32383+++	++0												2
4	Bound		InHost				+++29161+++	++0												2
5	Bound		InHost				+++23091+++	++0												2
6	Bound		InHost				+++22942+++	++0												2
7	Bound		InHost				+++20505+++	++0												2
8	Bound		InHost				+++19090+++	++0												2
9	Bound		InHost				+++14443+++	++0												2
10	Bound		InHost				+++13631+++	++0												2
11	Bound		InHost				+++7696+++	++0												2
12	Bound		InHost				+++7695+++	++0												2
13	Bound		InHost				+++3251+++	++0												2
14	Bound		InHost				+++1659+++	++0												2
15	Bound		InHost				+++49669+++	++0												2
16	Listen		InHost				+++49668+++	++0												2
17	Listen		InHost				+++49667+++	++0												2
18	Listen		InHost				+++49666+++	++0												2
19	Listen		InHost				+++49665+++	++0												2
20	Listen		InHost				+++49665+++	++0												2

- **Capturing memory using FTK manager.**





- **Generated hash**

```
Windows PowerShell
PS C:\Users\akshay> Get-FileHash C:\Users\akshay\Desktop\soc\memdump.mem -Algorithm SHA256

Algorithm      Hash
-----
SHA256         5F1DDAD16942907DCCA03E6418AD9F6470B2C86C7C5EF453E2CD6ADCC87512CE
C:\Users\akshay\Desktop\soc\m...
```

- **Chain of custody table**

Item	Description	Collected by	Date	Hash value
Memory dump	Win memory dump	SOC analyst	2026-05-14	5F1DDAD16942907DCCA03E6418AD9F6470B2C86C7C5EF453E2CD6ADCC87512CE

- **Summary**

- Evidence preservation is an essential component of cybersecurity investigations and incident response. Proper handling and storage of digital evidence help maintain data integrity, support forensic analysis, and improve incident investigation capabilities.
- The practical activity using Wazuh SIEM and Kali Linux provided hands-on experience in identifying, collecting, documenting, and preserving digital evidence during simulated security events.

- **Key Findings**

1. A functional SOC lab environment was successfully established using [Wazuh](#), enabling centralized monitoring, alert generation, and incident analysis.
2. Security monitoring activities demonstrated how SIEM platforms collect, normalize, and correlate logs from multiple sources to detect suspicious behavior.
3. Alert triage exercises showed the importance of prioritizing incidents based on severity, business impact, exploitability, and affected assets.



4. Practical attack simulations such as brute-force attempts and VSFTPD exploitation generated detectable events that were visible in the Wazuh dashboard.
5. MITRE ATT&CK mapping improved understanding of attacker tactics and techniques, helping classify incidents in a structured manner.
6. Log analysis exercises highlighted the usefulness of Windows Event IDs, Syslog entries, and network logs in identifying failed logins, unauthorized access attempts, and suspicious activities.
7. Dashboard creation and visualization improved the ability to identify trends such as top alert-generating IPs and repeated authentication failures.
8. Incident response documentation exercises emphasized the importance of maintaining timelines, investigation records, and remediation procedures during security incidents.

- **Conclusion**

The completion of these SOC and incident response tasks provided practical exposure to modern cybersecurity operations and monitoring workflows. The exercises strengthened understanding of SIEM-based monitoring, log analysis, alert management, threat intelligence integration, and incident response methodologies.

Through hands-on practice with Wazuh, simulated attacks, alert triage, dashboard creation, and evidence preservation activities, important operational SOC skills were developed. The tasks also highlighted the significance of documentation, forensic integrity, and standardized frameworks such as MITRE ATT&CK and NIST incident response guidelines.

Overall, the project successfully demonstrated the complete lifecycle of security monitoring and incident handling in a controlled lab environment, improving both theoretical understanding and practical cybersecurity capabilities.

- **References**

1. Wazuh Documentation
2. [MITRE ATT&CK Framework](#)
3. [NIST SP 800-61 Incident Handling Guide](#)
4. [SANS Incident Handler's Handbook](#)
5. [AlienVault OTX](#)
6. [VirusTotal](#)
7. [Wireshark Documentation](#)
8. [Snort Documentation](#)
9. [Osquery Documentation](#)
10. [Velociraptor Documentation](#)